Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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Abstract

Ocean source separators normally infer a changing background only from the mixture being separated. We study whether a causal forecast of the expected background, computed from prior soundscape measurements, improves separation. The proposed system encodes historical one-third-octave levels, predicts the current background spectrum, and injects either the forecast latent state or the predicted spectrum into the band tokens of BS-RoFormer. Forecast and separation losses are optimized jointly. We specify three controlled ablations, time-disjoint deployment splits, paired SI-SDR evaluation, and a real-time factor check. Because the available public product contains hourly spectral levels rather than phase-aligned audio, the present implementation uses random-phase background surrogates. The included smoke experiment validates the complete software and gradient path; it is not evidence of field performance. This distinction makes the prototype reproducible without overstating its current empirical scope.

1 Introduction

Passive acoustic backgrounds vary with shipping, weather, biology, and season. A separator that only observes the current mixture must estimate both the target and that background at once. Long-context soundscape forecasting offers an additional, strictly causal signal: what the background was expected to look like immediately before the mixture arrived. Our hypothesis is that this prior helps a separator reject predictable energy while retaining an unexpected injected source.

We combine the band-split RoPE Transformer (BS-RoFormer) architecture [1] with a causal recurrent forecast encoder. The contributions are: (i) a differentiable forecast–separation model; (ii) matched ablations for no forecast, latent conditioning, and spectrum conditioning; (iii) a leakage-audited deployment split; and (iv) an evaluation protocol that reports separation, forecast error, uncertainty, and computational usability.

2 Data and waveform construction

We use NOAA–Navy SanctSound hourly one-third-octave level (TOL) products [2]. MB03 deployments 01 and 02 form the training set, 03 is validation, and 04 is held-out test. A sample at hour t is eligible only when all H preceding hours and hour t are finite and exactly contiguous. Normalization statistics are fitted on the two training deployments only.

The public TOL product is an aggregate and cannot determine waveform phase. For each current spectrum S_t, we interpolate its dB envelope on FFT bins, draw a seeded random phase and Rayleigh-like magnitude jitter, invert the FFT, and normalize RMS. This gives background surrogate n_t. It preserves the coarse measured spectral envelope but not transients, phase, or within-hour events. It must not be described as recovered raw audio.

The injected signal s_t follows the controllable synthesis design developed in MINUS: harmonic partials plus band-limited noise, attack/release edges, and low-frequency amplitude modulation. Parameters and source-to-background SNR are redrawn deterministically for every sample. The mixture is $x_t = n_t + s_t$, and clipping is prevented by common rescaling.

3 Method

Let $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ be normalized historical TOL. A GRU produces

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \tag{1}$$

$$\widehat{S}_t = g_\theta(z_t). \tag{2}$$

Only observations ending at $t-1$ enter the encoder. The latent-conditioned model uses $c_t = z_t$; the spectrum-conditioned model uses $c_t = \widehat{S}_t$. The baseline supplies an all-zero condition and omits the forecaster.

BS-RoFormer converts mixture STFT bins into band tokens $X \in \mathbb{R}^{B\times T\times K\times D}$. Rather than concatenate an unstructured vector to audio, we learn a band-specific condition bias:

$$\widetilde{X}_{:,\tau,k,:} = X_{:,\tau,k,:} + [\mathrm{Linear}(\mathrm{LN}(c_t))]_{k,:}. \tag{3}$$

The conditioned tokens pass through the unchanged time/frequency Transformer stack and complex mask estimator. This location gives every time frame the same forecast prior while retaining frequency-specific effects.

The end-to-end objective is

$$\mathcal{L} = \mathcal{L}_{\text{wave}} + \tfrac{1}{2}\sum_r \mathcal{L}_{\text{complex STFT},r} + \lambda\|\widehat{S}_t - S_t\|_1, \tag{4}$$

with $\lambda = 0.2$ for conditioned variants. Gradients from both objectives reach the forecast encoder; this is joint training, not frozen feature export.

4 Experimental protocol

All variants use identical samples, synthesis seeds, optimizer schedule, model width, matched separator-core initialization, and checkpoint rule. The checkpoint with maximum validation SI-SDR improvement (SI-SDRi) is evaluated once on MB03_04. SI-SDR follows the scale-invariant projection form; improvement subtracts mixture SI-SDR. We also report residual-background SI-SDR, forecast MAE in dB, inference real-time factor (RTF), parameter count, and the fraction of paired test examples where conditioning beats the baseline. Metric uncertainty is a seeded paired bootstrap over test examples. BSS evaluation background is discussed by Vincent et al. [3].

Table 1: Smoke experiment on 128 TOL-derived surrogate examples.

Condition	SI-SDRi (dB)	Δ base (dB)	Forecast MAE (dB)	RTF
None (BS-RoFormer)	3.75	0	–	.0028
Predicted latent	3.44	-0.31	4.06	.0034
Predicted spectrum	4.15	+0.39	4.21	.0029

4.1 Smoke results

The small model was trained for 300 optimizer updates with 1,024 examples from each training deployment, 64 validation examples, and 128 fixed test examples. The unconditioned model reached 3.75 dB SI-SDRi. Latent conditioning reached 3.44 dB, a paired change of -0.31 dB (95% bootstrap interval $[-0.43, -0.20]$); spectrum conditioning reached 4.15 dB, a change of $+0.39$ dB ($[0.17, 0.61]$). The smoke result therefore supports the explicit predicted-spectrum interface but not the latent interface. This is evidence on surrogate waveforms only, not a field-audio claim. All variants had RTF below 0.004 on an RTX 3060, so compute is not the observed bottleneck. The conditioned models' forecast MAE was 4.06–4.21 dB. Full configuration and unrounded values are stored in `results/smoke_ablation.json`.

5 Usability criteria

The method is operationally usable only if its RTF is below one on the declared hardware, memory fits the deployment device, and performance remains stable across source SNR and time blocks. The scientific hypothesis is supported only if a conditioned variant has positive paired SI-SDRi gain over the unconditioned baseline and its 95% interval excludes zero on unseen, phase-aligned recordings. If forecast MAE improves while separation does not, the condition is either irrelevant or poorly injected. If oracle current spectra help but forecasts do not, forecasting is the bottleneck. These outcomes are explicitly falsifying, not post-hoc exceptions.

6 Limitations

Hourly TOL removes phase and within-hour temporal structure. Random-phase surrogates can make spectral priors artificially direct and underestimate real ocean complexity. The current test therefore establishes implementation correctness and controlled behavior only. The decisive experiment requires timestamp-aligned raw MB03 hydrophone audio: compute historical features without crossing the test boundary, mix held-out waveform clips with the same seeded sources, freeze all choices on MB03_03, and evaluate MB03_04 once. Multiple training seeds and SNR-stratified confidence intervals should then be reported.

7 Conclusion

FC-BSR supplies a concrete, leakage-resistant route from soundscape forecasting to source separation. Its frequency-specific conditioning path and three-way ablation isolate whether a compact forecast representation or an explicit predicted spectrum is more useful. The repository is ready for the full raw-audio experiment, while its present surrogate scope is stated plainly.

References

[1] Wei-Tsung Lu, Ju-Chiang Wang, Qiuqiang Kong, and Yun-Ning Hung. Music source separation with band-split RoPE transformer. *arXiv preprint arXiv:2309.02612*, 2023.

[2] NOAA Office of National Marine Sanctuaries and U.S. Navy. SanctSound raw passive acoustic data. NOAA National Centers for Environmental Information dataset, 2020.

[3] Emmanuel Vincent, Rémi Gribonval, and Cédric Févotte. Performance measurement in blind audio source separation. *IEEE Transactions on Audio, Speech, and Language Processing*, 14(4):1462–1469, 2006.